Mathematical and Computational Methods

Exercise Sheet 2: Complex numbers I: algebra and geometry

Tutorial route (▶, in this order, about 80 minutes): 1.3, 2.1(a,c), 2.3, 2.4, 2.5(a–c), 2.7, 2.10. Physics groups may replace 2.4(b) with 2.6. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill — powers of* i*)*. Simplify the powers of the imaginary unit

$$\mathrm{i}^3, \qquad \mathrm{i}^4, \qquad \mathrm{i}^{15}, \qquad \mathrm{i}^{2026}, \qquad \mathrm{i}^{-1}, \qquad \mathrm{i}^{-7}.$$

Exercise 1.2 *(⋆ drill — Cartesian arithmetic)*. With $z = 3 + 2\mathrm{i}$ and $w = 1 - 5\mathrm{i}$, find $z + w$, $z - w$, zw and z^2.

Exercise 1.3 *(▶ ⋆ drill — parts, conjugate and modulus)*. For $z = 24 - 7\mathrm{i}$ find: (a) $\operatorname{Re} z$ and $\operatorname{Im} z$; (b) $\overline{z}$; (c) $|z|$ and $z\,\overline{z}$; (d) $\operatorname{Im}(\mathrm{i}z)$.

Exercise 1.4 *(⋆ drill — division)*. Write $\dfrac{4 + 3\mathrm{i}}{1 + 2\mathrm{i}}$ in Cartesian form $a + b\,\mathrm{i}$.

Exercise 1.5 *(⋆ drill — quadratics over* $\mathbb{C}$*)*. Solve $x^2 + 1 = 0$ and $x^2 + 25 = 0$ over $\mathbb{C}$.

Exercise 1.6 *(⋆ drill — an Argand sketch)*. On a single Argand diagram, plot the points $z_1 = 3 + \mathrm{i}$, $z_2 = -2 + 2\mathrm{i}$ and $z_3 = -1 - 3\mathrm{i}$, together with the sum $z_1 + z_2$. Label each point.

Exercise 1.7 *(⋆ drill — polar and back)*. Convert to polar form $r(\cos\theta + \mathrm{i}\sin\theta)$ with principal argument: (a) $1 + \mathrm{i}$; (b) $2\mathrm{i}$; (c) -3. (d) Convert $z = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$ to Cartesian form.

Exercise 1.8 *(⋆ drill — the polar rules)*. Let $z = 3\big(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\big)$ and $w = 2\big(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\big)$. Using the polar rules, find zw and z/w in Cartesian form.

2 Tutorial problems

Exercise 2.1 *(▶ ⋆⋆ core — the conjugate and modulus rules)*. Let $z = 5 + \mathrm{i}$ and $w = 1 - \mathrm{i}$.

(a) Compute zw and z/w in Cartesian form.

(b) Verify the conjugate rule $\overline{zw} = \overline{z}\,\overline{w}$.

(c) Verify the modulus rule $|zw| = |z|\,|w|$.

Exercise 2.2 *(⋆⋆ core — two general proofs (maths))*. *(Maths stream — general proofs.)* Let $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$ with $a, b, c, d \in \mathbb{R}$.

(a) Prove the conjugate-of-a-product rule $\overline{zw} = \overline{z}\,\overline{w}$ directly from the definition $\overline{a + b\mathrm{i}} = a - b\mathrm{i}$.

(b) Using (a), prove that $\overline{1/w} = 1/\overline{w}$ for $w \neq 0$, and deduce that $\overline{z/w} = \overline{z}/\overline{w}$.

Exercise 2.3 *(▶ ⋆⋆ core — conjugate pairs).* Solve over $\mathbb{C}$, and confirm that the roots form a conjugate pair: (a) $x^2 - 4x + 20 = 0$, by completing the square; (b) $2x^2 + x + 1 = 0$, by the quadratic formula.

Exercise 2.4 *(▶ ⋆⋆ core — plotting and distance (data science)).* (a) Let $z = 3 + 2\mathrm{i}$ and $w = -1 + 2\mathrm{i}$. Plot z, w, $\overline{z}$, $z + w$, $z - w$ on an Argand diagram, and compute the distance $|z - w|$ between the points z and w.

(b) *(Data science — distance between records.)* Two data points in the plane are encoded as $p = 2 + 5\mathrm{i}$ and $q = 6 + 2\mathrm{i}$. Compute the Euclidean distance $|p - q|$ between them and the midpoint $\frac{1}{2}(p + q)$.

Exercise 2.5 *(▶ ⋆⋆ core — principal argument).* Convert each number to polar form, giving the *principal* argument $\operatorname{Arg} z \in (-\pi, \pi]$. Sketch each to fix the quadrant.

(a) $2\sqrt{3} + 2\mathrm{i}$; (b) $-1 + \mathrm{i}$; (c) $-2 - 2\sqrt{3}\,\mathrm{i}$; (d) $\sqrt{3} - \mathrm{i}$; (e) -5.

Exercise 2.6 *(⋆⋆ core — AC impedance (physics)).* *(Physics — AC circuits.)* A series circuit has complex impedance $Z = 4 + 3\mathrm{i}$ (in ohms). It is driven by a voltage $V = 10$ (volts, taken as a real reference).

(a) Find $|Z|$ and the phase angle $\operatorname{Arg} Z$.

(b) Compute the current $I = V/Z$ in Cartesian form and its magnitude $|I|$.

Exercise 2.7 *(▶ ⋆⋆ core — polar product and quotient).* Let $z = \sqrt{2}\left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right)$ and $w = 2\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$. Find (a) zw and (b) z/w, giving each in polar form with the *principal* argument and in Cartesian form.

Exercise 2.8 *(⋆⋆ core — multiplication by i (with phasors)).* (a) For $z = 4 + \mathrm{i}$, compute $\mathrm{i}z$ and confirm geometrically that multiplying by i is a quarter-turn anticlockwise that preserves length.

(b) Which single complex number rotates a point by $\frac{\pi}{2}$ anticlockwise *and* doubles its distance from the origin?

(c) Apply a clockwise quarter-turn to $3 + \mathrm{i}$ (i.e. multiply by the number of modulus 1 and argument $-\frac{\pi}{2}$).

(d) *(Physics — phasors.)* A voltage phasor $V_1 = 6 + 8\mathrm{i}$ is joined by a second phasor V_2 of equal amplitude turned a quarter-turn anticlockwise, i.e. $V_2 = \mathrm{i}V_1$. Find V_2, the resultant $V = V_1 + V_2$ in Cartesian form, and its amplitude $|V|$.

Exercise 2.9 *(⋆⋆ core — true or false).* *(Conceptual — true or false, with a one-line reason.)*

(a) For every $z \in \mathbb{C}$, $z^2 \geq 0$.

(b) $|z| = |\overline{z}|$ for all z.

(c) If $z + \overline{z} = 0$ then z is purely imaginary or zero.

(d) $\operatorname{Re}(zw) = \operatorname{Re}(z)\,\operatorname{Re}(w)$ for all z, w.

(e) There is an ordering $<$ on $\mathbb{C}$ compatible with its arithmetic — that is, one respecting $+$ and $\times$ (if $x > 0$ and $y > 0$ then $x + y > 0$ and $xy > 0$), so that every nonzero square is positive.

Exercise 2.10 *(▶ ⋆ ⋆ ⋆ challenge — an equilateral triangle).* *(Rotation/geometry.)* Two vertices of an equilateral triangle lie at 0 and 2 on the real axis. Use multiplication by a complex number of modulus 1 to find the two possible positions of the third vertex.

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core — solve for real x, y)*. Find the real numbers x, y satisfying $(x+y\,\mathrm{i})(3-\mathrm{i}) = 7+\mathrm{i}$.

Exercise 3.2 *(⋆⋆ core — division)*. Simplify to Cartesian form: (a) $\dfrac{1+2\mathrm{i}}{3-4\mathrm{i}}$; (b) $\dfrac{\mathrm{i}}{1+\mathrm{i}}$.

Exercise 3.3 *(⋆⋆ core — a real cubic from one root)*. Given that $1+2\mathrm{i}$ is a root of $x^3-4x^2+9x-10=0$ (real coefficients), find the other two roots.

Exercise 3.4 *(⋆⋆ core — loci and a region in the plane)*. Describe geometrically, and sketch, the set of points $z \in \mathbb{C}$ satisfying

(a) $|z| = 2$; (b) $|z-1| = 3$; (c) $|z-\mathrm{i}| = |z+\mathrm{i}|$; (d) $|z-1| < 2$.

For (d), shade the region and show clearly whether its boundary is included.

Exercise 3.5 *(⋆⋆ core — one product, two ways)*. Compute the product $(1+\mathrm{i})(\sqrt{3}+\mathrm{i})$ in two ways — by Cartesian expansion and by the polar rule — and hence write down its modulus and argument.

Exercise 3.6 *(⋆⋆ core — modulus and argument round-up)*. Suppose $|z| = 2$, $\operatorname{Arg} z = \frac{\pi}{3}$, $|w| = 3$, $\operatorname{Arg} w = \frac{\pi}{4}$. Find the modulus and argument of (a) zw, (b) z/w, (c) z^2 and (d) $\overline{z}$; (e) convert z and z^2 to Cartesian form.

Exercise 3.7 *(⋆⋆⋆ challenge — the parallelogram law)*. Prove the *parallelogram law*: for all $z, w \in \mathbb{C}$,

$$|z+w|^2 + |z-w|^2 = 2\,|z|^2 + 2\,|w|^2 .$$

Exercise 3.8 *(⋆⋆⋆ challenge — a square root in Cartesian form)*. Find all $z = x + y\,\mathrm{i}$ (with $x, y \in \mathbb{R}$) such that $z^2 = 3 + 4\mathrm{i}$, working directly in Cartesian form.